

Preconditioner Benchmarks (Deparis-Grandperrin-Quarteroni style campaign)

Naming convention: P = Preconditioner. These tests are not physical validation runs; they compare the linear solver and preconditioner behaviour on representative Navier-Stokes saddle-point systems.

Common baseline	Value / convention
Base problem	2D flow past cylinder, mesh = L0, constant parabolic inlet, Re = 100, nu = 1e-3, theta = 0.6
Time window	Run to T = 2.0. Compute performance statistics only on t >= 1.0; t < 1.0 is warm-up and excluded from averages.
Linear/nonlinear setup	Oseen linearization, nonlinear_iterations = 1. Outer GMRES relative tolerance = 1e-6, absolute tolerance = 1e-10, restart = 300, max iterations = 10000.
Default parallelism	mpi_ranks = 4, except in P8 where ranks are swept over 1, 2, 4, 8.
Preconditioners	simple, pcd, yosida, block_triangular are included in P1, P2, P3, P4, P7 and P8. P5 is SIMPLE-only; P6 is PCD-only.
Output policy	Write solution output = false for performance benchmarks; otherwise visualization I/O contaminates timings.
Core metrics	GMRES iterations, setup time, linear solve / preconditioned iteration time, time per GMRES iteration, total step time, runtime, failures. For P8 also speedup, efficiency and iteration growth.

ID	Previous ID	Benchmark	Purpose	Fixed setup	Sweep	Preconditioners	Main reported quantities
P0	P0	Preconditioner sanity run	Check that all preconditioners complete a short run.	T = 0.2-0.5, L0, dt = 0.001, Re = 100.	-	simple, pcd, yosida, block_triangular	Convergence flag, GMRES failures, basic timings.
P1	P1	Baseline preconditioner comparison	Main direct comparison at fixed problem parameters.	T = 2, statistics t >= 1, L0, dt = 0.001, Re = 100, mpi = 4.	-	simple, pcd, yosida, block_triangular	GMRES mean/max, setup time, solve time, time per iteration, runtime, failures.
P2	P2	Timestep sensitivity	Measure how preconditioners react to changing dt.	T = 2, statistics t >= 1, L0, Re = 100, mpi = 4.	dt = 0.01, 0.005, 0.0025, 0.001	simple, pcd, yosida, block_triangular	GMRES vs dt, setup time, solve time, runtime.
P3	P3	Mesh sensitivity / optimality	Assess whether iterations remain controlled as DoFs increase.	T = 2, statistics t >= 1, dt = 0.001, Re = 100, mpi = 4.	mesh = L0, L1; L2 optional if affordable	simple, pcd, yosida, block_triangular	Velocity DoFs, pressure DoFs, total DoFs, cells, GMRES, setup, solve, runtime.
P4	P4	Reynolds / viscosity sensitivity	Assess robustness with respect to physical parameters.	T = 2, statistics t >= 1, L0, dt = 0.001, mpi = 4.	Re = 20, 100, 200 with nu = 0.005, 0.001, 0.0005	simple, pcd, yosida, block_triangular	GMRES vs nu/Re, failures, solve time, runtime.
P5	P5	SIMPLE relaxation sweep	Tune the SIMPLE pressure relaxation parameter alpha.	T = 2, statistics t >= 1, L0, dt = 0.001, Re = 100, mpi = 4.	alpha = 0.2, 0.3, 0.5, 0.7, 0.9, 1.0	simple only	alpha, GMRES mean/max, solve time, runtime, best alpha.
P6	P6	PCD inner tolerance sweep	Find the best trade-off between inner PCD accuracy and total time.	T = 2, statistics t >= 1, L0, dt = 0.001, Re = 100, mpi = 4.	pcd_pressure_relative_tolerance = 1e-1, 1e-2, 1e-3, 1e-4	pcd only	Outer GMRES, inner iterations if available, setup time, solve time, runtime.
P7	P10	Cost breakdown	Separate assembly, preconditioner setup and linear solve cost.	T = 2, statistics t >= 1, L0, dt = 0.001, Re = 100, mpi = 4.	-	simple, pcd, yosida, block_triangular	Assembly time, preconditioner setup time, linear solve time, total step time, time per GMRES iteration.
P8	P9	MPI strong scaling	Measure strong scaling and preconditioner scalability.	T = 2, statistics t >= 1, L0, dt = 0.001, Re = 100.	mpi_ranks = 1, 2, 4, 8	simple, pcd, yosida, block_triangular	Runtime, mean step time, speedup, efficiency, GMRES mean, iteration growth.
P9	P11	Optional 3D preconditioner test	Qualitative 3D solver/preconditioner comparison.	3D_L0, T = 1-2, dt = 0.005 or 0.001, Re = 100, mpi = 4.	Optional	simple, pcd, block_triangular; yosida only if affordable	GMRES, setup time, solve time, runtime, stable.

Paper-aligned reporting: Deparis-Grandperrin-Quarteroni report iteration counts, setup time and preconditioned iteration / solve time, and use sensitivity to dt, viscosity, mesh size and parallel ranks to assess efficiency, robustness, optimality and scalability. Inner iterations are useful to log when available, especially for PCD, but the primary comparison should remain outer GMRES + setup/solve timings.